Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching

Project code: PKV Group: GFA26SE111

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Table of Contents

I. Record of Changes 3
II. Testing Documentation 3
1. Scope of Testing 3
2. Test Strategy 4
2.1 Testing Types 4
2.2 Test Levels 5
2.3 Supporting Tools 5
3. Test Plan 6
3.1 Human Resources 6
3.2 Test Environment 6
3.3 Test Milestones 6
4. Test Cases 6
4.1 Requirement Traceability 7
5. Test Reports 8
5.1 Evidence 8
5.2 Unit Test Results 8
5.3 System and Integration Test Results 9
5.4 Defects 9
5.5 Analysis and Next Actions 10

I. Record of Changes

Date	A* M, D	In charge	Change Description
27/10/2026	A	Nguyễn Gia Huy	v0.1 – Test formula, workflows and demo scenarios prepared for the Review 2 commitment (BM04).
06/11/2026	A	Nguyễn Văn Thành Đạt	v0.2 – First test documentation: 116 planned cases, 36 unit checks, no results.
11/11/2026	M	Nguyễn Gia Huy	v1.0 – Rewritten: scope with levels, entry and exit criteria; test types with objective, technique and completion criteria; tools with vendor and version; dated test milestones; 114 concrete test cases mapped to FR-01–FR-33; real results of 09–11/11/2026 (515 EditMode tests passed; 49 passed, 3 failed, 62 pending system cases); 18 defects; analysis against the release criteria.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

The target of test is PickleVerse release 1.0 as specified in Report 3: the ten features FE-01 to FE-10 with their functional requirements FR-01 to FR-33, the business rules BR-01 to BR-51, and the non-functional requirements NFR-01 to NFR-12. Testing covers the Unity game client (Windows build and Unity Editor), the dedicated game server, the Web API with its workers, the AI coaching service and the administration web application deployed on the staging server.

In scope. All screens and non-screen functions of Report 3, the gameplay rules and physics parameters of Report 4 §3.2, online matchmaking and dedicated play, data integrity of rewards and results, role-based access, and the measurable quality attributes NFR-01 to NFR-12.

Out of scope. iOS builds and real-money payment (Report 1, LI-02 and LI-06); load beyond the API capacity target, because the staging deployment has one dedicated-server slot (LI-07); third-party services themselves (Photon Cloud, Gmail SMTP), which are only tested through their integration points; and the tier model's accuracy until enough expert-labelled data exists (NFR-06).

Table 1. Test levels: responsibility, inputs, focus and exit criteria

Level	In charge	Inputs and time	Focus	Acceptance (exit) criteria
Unit	Each module owner; review by Trần Nguyễn Hiếu Thuận	Source code, BR-xx; before every merge	Rules, scoring, Elo, input decoding, economy, offline store, matchmaking policies (Unity EditMode, pytest)	All unit tests pass; every public rule method has at least one normal, abnormal and boundary case
Integration	Nguyễn Văn Thành Đạt	API build, Database.sql seed, PostgreSQL and Redis; each package milestone	API contracts, database writes, dedicated tickets and leases, training sync receipts, role and scope checks	All IntegrationChecks pass on a fresh database; no contract difference between Unity, API and AI service
System	Trần Nguyễn Hiếu Thuận with the feature owners	Staging deployment, test accounts and seed data; Rounds 1–3 (§3.3)	Every FR through the user interface, state transitions, NFR measurements	≥ 90% of system cases passed, all FRs covered by a passed case, no open Critical/High defect
Acceptance	Huỳnh Trung Kiên with the supervisor	Release candidate, demo scenarios W01–W04	End-to-end workflows of the four members	All four workflows pass in front of the supervisor

Constraints and assumptions.

- The staging server runs one dedicated-server container, so online cases that need two simultaneous matches cannot be executed.
- GitHub Actions has no Unity licence; Unity tests are run with the Unity CLI on a team PC and the result file is stored with the commit hash.
- Tests that need several players (rooms, friends, reports) require two or more accounts and devices at the same time.
- A dedicated test mailbox and seed accounts (for example a player at level 24 or with 9 passed lessons) must be added to Database.sql before Round 2.
- The Photon free plan limits concurrent users; group playtests are scheduled so that the limit is not reached.

2. Test Strategy

2.1 Testing Types

Table 2. Testing types

Type	Objective	Technique	Completion criteria
Functional	Verify every FR and BR	Black-box: equivalence partitions and boundary values from BR-xx; decision tables for scoring and Elo	Each FR has ≥ 1 passed case; all Main cases passed
Gameplay rules and physics	Verify serve, faults, scoring, contact timing and bounce	Automated EditMode tests on the engine-independent rule classes; scripted rallies	All rule tests pass; every fault type has a test
User interface and usability	Verify layout, messages and learnability	Checklist per screen against Report 3 §3.1 and the message list; moderated usability test (NFR-01)	No overlapping or truncated elements; NFR-01 met
Integration / API contract	Verify API, database and service contracts	IntegrationChecks console harness on a fresh database; shared JSON fixtures in contracts/	All checks pass on the release commit
Security	Verify authentication, authorisation and secret handling	Negative API tests with wrong roles and scopes, token expiry, rate limit; secret scan	NFR-09 checklist satisfied; no secret in the repository
Performance and load	Verify NFR-02 to NFR-05	Unity Profiler on reference devices; Fusion RTT statistics; k6 load test on the API	All measured values within the NFR limits
Network resilience	Verify reconnection and interruption	Disconnect a client for 20 s and 60 s; drop the dedicated server heartbeat	Behaviour matches BR-28 and BR-29

Type	Objective	Technique	Completion criteria
Compatibility	Verify target platforms	Install and smoke test on Windows 10/11 and Android 8+ (ARM64); admin web on Chrome and Edge	NFR-10 met
AI quality	Verify coaching findings and the gated model	Holdout evaluation against expert labels	NFR-06 thresholds met, otherwise the model stays disabled
Exploratory playtest and balance	Find defects that scripted cases miss; tune bot difficulty	Time-boxed sessions with a charter; win rate of players of known level against each bot level	Each session logged; balance targets agreed with the supervisor
Regression	Keep fixed defects fixed	Re-run the automated suites and the cases linked to closed defects in every round	No reopened defect

2.2 Test Levels

Table 3. Test types per level

Type of Tests	Unit	Integration	System	Acceptance
Functional	X	X	X	X
Gameplay rules and physics	X		X	X
User interface and usability			X	X
Integration / API contract		X		
Security	X	X	X	
Performance and load		X	X	
Network resilience	X		X	
Compatibility			X	X
AI quality	X		X	
Exploratory playtest and balance			X	X
Regression	X	X	X	

2.3 Supporting Tools

Table 4. Supporting tools

Purpose	Tool	Vendor/In-house	Version
Unity unit tests (EditMode, NUnit)	Unity Test Framework	Unity Technologies	1.8.0
Run tests and capture screens from the command line	Unity CLI with the Pipeline package	Unity Technologies	CLI 1.0.0-beta.17, Pipeline 0.9.0-exp.1
Python unit tests	pytest (with httpx)	Open source	8.x
API and database integration checks	IntegrationChecks console harness	In-house	main @ 7f8ee32
Manual API calls	Swagger UI	Swashbuckle (open source)	10.3.1
API load test (NFR-04)	k6	Grafana Labs (open source)	latest stable, planned
Frame-rate measurement	Unity Profiler	Unity Technologies	6000.6.2f1
Network statistics	Photon Fusion statistics and dashboard	Exit Games	Fusion 2.1.3
Test case and result management	Microsoft Excel (Report5 workbooks)	Microsoft	Microsoft 365
Continuous integration	GitHub Actions (validate job)	GitHub	Hosted runner

3. Test Plan

3.1 Human Resources

Table 5. Human resources

Worker/Doer	Role	Specific Responsibilities/Comments
Trần Nguyễn Hiếu Thuận	Test lead	Owns the test plan and workbooks; runs system rounds; tracks defects; tests FE-01 and FE-06 screens; workflow W02
Nguyễn Gia Huy	Tester, gameplay owner	Unit tests of rules, physics, training and bot; playtests; runs the Unity EditMode suite; workflow W01
Nguyễn Văn Thành Đạt	Tester, backend and network owner	IntegrationChecks, API security and load tests, online and dedicated cases; workflow W03
Huỳnh Trung Kiên	Tester, leader	Admin web cases, acceptance session with the supervisor, release decision; workflow W04
Mr. Lại Đức Hùng	Supervisor	Witnesses the acceptance scenarios and approves the release criteria

3.2 Test Environment

Table 6. Test environment

Purpose	Tool	Provider	Version
Staging server	Ubuntu with Docker Engine and Docker Compose (API, AI service, dedicated server, PostgreSQL, Redis, S3 storage, web)	Team VPS tad-min.io.vn	Ubuntu 24.04; images of main
Game client under test	Unity Editor and Windows build	Unity Technologies	6000.6.2f1
Test PC	Windows PC with keyboard and mouse	Team member	Windows 11 Home
Android device	Android phone (ARM64) for NFR-03 and NFR-10	Team member	Android 8.0 or later (to be recorded in Round 2)
Real-time service	Photon Cloud	Exit Games	Fusion 2.1.3
Database and cache	PostgreSQL, Redis	Docker images (pinned digests)	16 / 7.4
Admin web browser	Google Chrome, Microsoft Edge	Google, Microsoft	Latest stable

3.3 Test Milestones

Table 7. Test milestones

Milestone Task	Start Date	End Date
Round 1 – automated suites and first system pass on staging	09/11/2026	13/11/2026
Prepare seed accounts, test mailbox and analytics seed	16/11/2026	18/11/2026
Round 2 – all system cases; fix verification of D-01 to D-18	16/11/2026	27/11/2026
Usability test with 10 students (NFR-01)	23/11/2026	27/11/2026
Load, latency and frame-rate measurements (NFR-02 to NFR-05)	23/11/2026	29/11/2026
Uptime probe for NFR-07 (14 days)	30/11/2026	13/12/2026
Round 3 – regression on the release candidate	30/11/2026	06/12/2026
Acceptance test W01–W04 with the supervisor	07/12/2026	11/12/2026

4. Test Cases

Unit test cases are documented in **Report5_Unit Test.xlsx** as decision tables (conditions, confirmations, type N/A/B and result for each UTCID) for ten core functions, each mapped to the NUnit test that implements it. Integration, system and acceptance test cases are documented in **Report5_Test Report.xlsx**, one sheet per feature group, with ID, description, procedure with concrete data, expected result, pre-conditions, three test rounds and a note that names the evidence or the reason why a case is pending.

The number of cases follows the Review 2 formula: 18 committed features × 3 cases (54) + 4 workflows + 42 state transitions (three per state machine SM-01 to SM-14) + 12 NFRs = 112 cases. Five cases are

added: two for the expert review tool (FE-10), which is not in the Review 2 feature list, and three for FR-03 to FR-05, which the Review 2 feature F01 did not detail, giving **117 cases**.

Table 8. Test cases by sheet and level

Sheet	Scope	Cases	Unit	Integration	System	Acceptance	Automated
FE01 Account	FE-01 Account and onboarding	6	1	0	5	0	1
FE02 Training	FE-02 Tutorial, training and offline	9	4	0	5	0	4
FE03 Bot	FE-03 Bot practice	3	2	0	1	0	2
FE04 Online	FE-04 Online matches	12	8	0	4	0	8
FE05 Coaching	FE-05 AI coaching	3	1	0	2	0	1
FE06 Progress	FE-06 Progress, rewards and shop	6	4	0	2	0	4
FE07 Social	FE-07 History, social and ranking	6	2	1	3	0	2
FE08-09 Admin	FE-08/FE-09 Moderation and administration	12	1	1	10	0	1
FE10 Expert	FE-10 Expert review	2	0	0	2	0	0
Workflows	End-to-end workflows W01–W04	4	0	0	0	4	0
States	State transitions SM-01 to SM-14	42	11	27	4	0	11
NFR	Non-functional requirements	12	0	0	12	0	0
Total		**117**	**34**	**29**	**50**	**4**	**34**

4.1 Requirement Traceability

Table 9. Traceability from functional requirements to test cases

Requirement	Test cases	Cases	Passed	Coverage status
FR-01	TC-F01-01, TC-F01-03, TC-W01	3	1	Covered
FR-02	TC-F01-01, TC-F01-02	2	1	Covered
FR-03	TC-F01-04	1	0	Not yet covered
FR-04	TC-F01-05	1	0	Not yet covered
FR-05	TC-F01-06	1	1	Covered
FR-06	TC-F02-01, TC-F02-02, TC-W01	3	1	Covered
FR-07	TC-F03-01, TC-F03-03	2	2	Covered
FR-08	TC-F03-01, TC-F03-02, TC-W01	3	2	Covered
FR-09	TC-F14-01, TC-F14-02, TC-F14-03, TC-F06-03, TC-W01	5	2	Covered
FR-10	TC-F11-01	1	1	Covered
FR-11	TC-F02-03	1	1	Covered
FR-12	TC-F05-01, TC-F05-03, TC-F09-02	3	3	Covered
FR-13	TC-F05-01, TC-F05-02, TC-F05-03, TC-W02	4	4	Covered
FR-14	TC-F07-01, TC-F07-02, TC-F07-03, TC-F10-02, TC-W03	5	4	Covered
FR-15	TC-F08-01, TC-F08-02, TC-F08-03	3	2	Covered
FR-16	TC-F09-01, TC-F09-02, TC-W03	3	2	Covered
FR-17	TC-F09-03	1	0	Not yet covered
FR-18	TC-F09-01, TC-F10-01, TC-F10-02, TC-F10-03, TC-W02, TC-W03	6	5	Covered
FR-19	TC-F06-01, TC-F06-02, TC-F06-03, TC-W02	4	3	Covered
FR-20	TC-F04-03	1	0	Not yet covered
FR-21	TC-F13-01, TC-F13-02, TC-F13-03	3	3	Covered
FR-22	TC-F04-01	1	1	Covered
FR-23	TC-F04-02	1	1	Covered
FR-24	TC-F11-02, TC-F11-03	2	1	Covered
FR-25	TC-F12-01, TC-F12-02	2	1	Covered
FR-26	TC-F12-03	1	1	Covered
FR-27	TC-F17-01, TC-W04	2	0	Not yet covered
FR-28	TC-F17-03, TC-W04	2	0	Not yet covered
FR-29	TC-F15-01, TC-F15-02, TC-F15-03	3	0	Not yet covered
FR-30	TC-F17-01, TC-W04	2	0	Not yet covered

Requirement	Test cases	Cases	Passed	Coverage status
FR-31	TC-F16-01, TC-F16-02, TC-F16-03	3	1	Covered
FR-32	TC-F17-02, TC-F18-01, TC-F18-02, TC-F18-03	4	0	Not yet covered
FR-33	TC-X01-01, TC-X01-02	2	0	Not yet covered

23 of the 33 functional requirements are covered by at least one passed case after Round 1. NFR-01 to NFR-12 are traced one-to-one to TC-N01 to TC-N12.

5. Test Reports

5.1 Evidence

Table 10. Test evidence

ID	Run	Date	Build / commit	Result	Location
EV-01	Unity EditMode suite through the Unity CLI (Pipeline test run)	11/11/2026	main @ 7f8ee32	515 tests, 515 passed, 0 failed, 6.19 s	Doc/Project_report/Report/evidence/editmode_results_20261111.json
EV-02	Playtest on the staging server (offline training, signed-in menu, bot match, AI coaching, ranked 1v1 with bot fallback)	10/11/2026	main @ 82f6a7a client; staging images of 06/11/2026	See Report5_Test Report.xlsx	screenshots/01–28, evidence/playtest_log_20261110.md
EV-03	UI check of the register, verification, password-reset screens and the tutorial	11/11/2026	main @ 7f8ee32	See TC-F01-03, TC-F02-01	screenshots/29–34
EV-04	Match history of the team test accounts (two ranked 1v1 attempts)	09/11/2026	staging	Both interrupted	screenshots/12_history.png
EV-05	Team-recorded IntegrationChecks on disposable PostgreSQL databases (12 areas)	06/11/2026	df5f671	Reported as passed; logs not in the repository, so not counted	Review3_Evidence.md (file names and SHA-256)

5.2 Unit Test Results

Table 11. Unity EditMode results by assembly (EV-01)

Test assembly	Tests	Passed	Failed
Audio.Tests	23	23	0
FrontEnd.Tests	4	4	0
HybridBot.Tests	14	14	0
LocalMatch.Tests	157	157	0
Multiplayer.Tests	101	101	0
Progression.Tests	125	125	0
Replays.Tests	16	16	0
Services.Tests	31	31	0
Shop.Tests	33	33	0
Social.Tests	11	11	0
Total	**515**	**515**	**0**

The ten decision tables of Report5_Unit Test.xlsx contain 66 unit test cases (normal, abnormal and boundary), all passed in EV-01. The other automated suites were not executed for this report: the 22 pytest tests of the AI service need FastAPI and scikit-learn, which are not installed on the test PC, and the IntegrationChecks need restored NuGet packages and a PostgreSQL/Redis instance. The team recorded passing IntegrationChecks runs on 06/11/2026 (EV-05), but their logs are not in the repository, so they are not counted here. Both suites run in the GitHub Actions validate job, whose latest result must be attached in Round 2. Code coverage has not been measured yet; the Unity Code Coverage package will be enabled in Round 2.

5.3 System and Integration Test Results

Table 12. Round 1 results by module

Module	Cases	Passed	Failed	Pending
FE-01 Account and onboarding	6	3	0	3
FE-02 Tutorial, training and offline	9	7	0	2
FE-03 Bot practice	3	3	0	0
FE-04 Online matches	12	8	2	2
FE-05 AI coaching	3	2	1	0
FE-06 Progress, rewards and shop	6	5	0	1
FE-07 History, social and ranking	6	4	0	2
FE-08/FE-09 Moderation and administration	12	1	0	11
FE-10 Expert review	2	0	0	2
End-to-end workflows W01–W04	4	2	0	2
State transitions SM-01 to SM-14	42	15	0	27
Non-functional requirements	12	0	0	12
Total	**117**	**50**	**3**	**64**

Table 13. Release criteria after Round 1

Metric	Value	Target	Status
Test coverage (executed / total)	45.3% (53 / 117)	100% before release	Not met
Pass rate of executed cases	94.3% (50 / 53)	≥ 90%	Met for executed cases
Test successful coverage (passed / total)	42.7%	≥ 90%	Not met
Automated cases	29.1% (34 / 117)	≥ 50%, run in CI	Not met
Functional requirements covered by a passed case	23 / 33	33 / 33	Not met
Open Critical / High defects	0 / 3 (D-09, D-17, D-18)	0 / 0	Not met

5.4 Defects

Table 14. Defects found in Round 1 (all Open)

ID	Summary	Type	Severity	Found by	Owner
D-01	Offline banner overlaps the titles of Progress, the training HUD and the lesson list	UI	Low	TC-F14-01	Thuận
D-02	Offline: the analysis area stays at "Đang phân tích…" forever	Functional	Medium	TC-F06-03	Đạt
D-03	Key-hint box of the training HUD covers the player character	UI	Low	TC-F03-01	Huy
D-04	"Quyết định & Khiếu nại" button overlaps the PLAY button and the profile panel	UI	Medium	TC-F01-02	Thuận
D-05	Fourth bot difficulty shows the technical name "BhTree"	UI text	Low	TC-F05-01	Huy
D-06	Bot card says "Không cần kết nối mạng" but bot matches are disabled offline	UI text	Low	TC-F14-01	Thuận
D-07	Male/female choice is not applied in bot match, training and tutorial (always the female character)	Functional	Medium	TC-F05-01	Huy
D-08	New players start at Elo 100 while the admin reset-Elo default is 2000	Business rule	Medium	TC-F10-01	Đạt
D-09	Ranked 1v1 between two humans ended "Gián đoạn" twice, without replay	Functional	High	TC-F09-01	Đạt
D-10	"Xem phân tích" button on the match result is oversized and covers the centre of the screen	UI	Low	TC-F05-01	Thuận
D-11	Coaching popup mixes session-wide (2/86) and per-technique (1/16) figures	Usability	Low	TC-F06-01	Đạt
D-12	Elite Token has no earning source, so GEMSTONE can only be granted at level 25	Requirement	Medium	TC-F04-01	Kiên
D-13	Hard bot is beaten 2–0 by the auto-player with short rallies; balance not validated	Balance	Medium	TC-F05-01	Huy
D-14	After a normal ranked result the lobby shows "Mất kết nối Photon."	UI message	Low	TC-W03	Đạt
D-15	Ranked bot offered about 160 s after the search started instead of 180 s	Functional	Low	TC-F07-01	Đạt
D-16	Sync-UnityTests.ps1 fails to parse in Windows PowerShell 5.1 (UTF-8 file without BOM)	Tooling	Medium	Unit test run	Thuận

ID	Summary	Type	Severity	Found by	Owner
D-17	Development credentials committed in appsettings.json and a fallback JWT key in Program.cs	Security	High	TC-N09	Đạt
D-18	Staging API and admin web are served over HTTP instead of HTTPS	Security	High	TC-N09	Đạt

Round 1 found 18 defects: 3 High, 7 Medium and 8 Low. Severity follows Report 3 NFR-08: Critical means loss of player data or inability to sign in or play; High means a main workflow or a security requirement fails; Medium means a requirement is partly wrong or a rule is inconsistent; Low means a cosmetic or wording problem. The defect life cycle (Open → Fixed → Verified → Closed) is tracked in PickleVerse_Project_Tracking.xlsx.

5.5 Analysis and Next Actions

- **Engine-level quality is good.** All 515 EditMode tests pass, and every system case that could be executed for the single-player features (training, bot match, coaching, progress screens) passed, apart from the offline analysis status (D-02).
- **Online play between humans is the main risk.** The only human-versus-human ranked attempts (EV-04) were interrupted (D-09), which also blocks the reconnection, report and appeal cases. Reproducing D-09 with server logs is the first task of Round 2 (owner: Đạt, by 18/11/2026).
- **Security requirements are not yet met** because of committed credentials and HTTP on staging (D-17, D-18). Both must be fixed before the acceptance test.
- **Coverage is limited by test data, not by tooling.** 62 cases are pending mainly because seed accounts, a test mailbox and a second player were not available. These will be added to Database.sql before Round 2 (owner: Thuận, by 18/11/2026).
- **Automation is below the 50% target and not in CI.** 33 of 114 cases are automated. The EditMode suite will be run in CI with a GameCI runner or, if no licence is available, by a scheduled Unity CLI run whose result file is committed; the pytest and IntegrationChecks results of the validate job will be linked to their test case IDs.